

Project of survival Analysis

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Introduction

Dataset description

This dataset comes from an experimental study investigating whether invasive (non-native) populations of *Ulmus pumila* (Siberian elm) differ in their germination dynamics compared to native populations. The central ecological question is whether non-native populations exhibit faster or more successful germination, which may contribute to their invasion success.

Seeds were collected from two geographic ranges:

- **China**, representing the native range,
- **United States**, representing the non-native range.

Germination experiments were conducted under two controlled temperature treatments:

- **20°C** (moderate temperature),
- **30°C** (warm temperature).

For each seed, the time to germination was recorded. Seeds that did not germinate by the end of the observation period were treated as **right-censored observations**.

Research question

Do non-native populations of *Ulmus pumila* germinate faster or more successfully than native populations, and how is this affected by temperature?

Recorded variables

- **seed**: Index of the seed within each experimental replicate (values range from 1 to 20). This variable is *not* a unique identifier, as the same seed index appears across multiple replicates.
- **Range**: Geographic origin of the seed (**China** = native, **USA** = non-native).
- **Population**: Specific population within each geographic range.
- **Replicate**: Experimental replicate or batch to which the seed belongs.
- **Germination__day**: Day on which germination occurred (discrete time variable).
- **Status**: Censoring indicator (1 = seed germinated, 0 = seed did not germinate by the end of the study).
- **Temperature__treatment**: Temperature condition under which the seed was incubated (20°C or 30°C).

Statistical framework

Because germination time is observed as a discrete time-to-event variable with right censoring, the data naturally lend themselves to **survival analysis** methods, such as Kaplan–Meier estimators, log-rank tests, and Cox proportional hazards models.

Part A - Descriptive statistics

Now that we know what is the dataset, we will now explore the dataset. Let's see how many species there are within each geographic origin.

Table 1: Geographic Origin by Population

	AF2	AF5	BJ	CD2	CX1	CX2	CX3	CX4	CX5	GJ1	IM	SF4	SLC1	WP34	Total
CHINA	0	0	145	0	236	289	281	258	269	0	276	0	0	0	1752
U.S.	296	268	0	272	0	0	0	0	0	260	0	267	262	250	1875

We can see from table 1 that the populations are more less symetrically distributed and that there are 7 populations of Native origin and 7 populations of non native origin. Let us now, be interested in the descriptive statistics of our survival variable Germination_day (variable of interest).

Table 2: Summary Statistics of Germination Day by Geographic Origin and Temperature Treatment

Range	Temperature	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
CHINA	20C	2.0	5.0	5.0	6.0	7.0	14.0
U.S.	20C	2.0	5.0	5.0	4.9	5.0	14.0
CHINA	30C	2.0	2.0	2.0	3.6	5.0	14.0
U.S.	30C	2.0	2.0	2.0	2.7	2.0	14.0

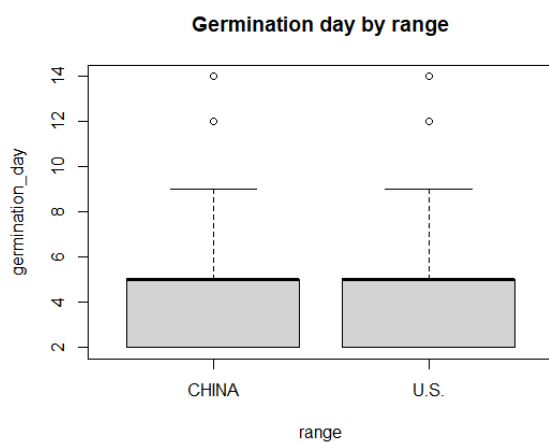


Figure 1: Germination Day vs Range

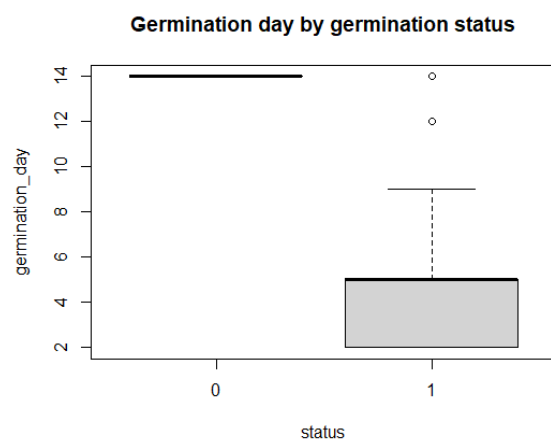


Figure 2: Germination Day vs Status

Table 2 shows us the descriptive statistics of our Germination_Day with respect to Range and Temper-

ature_Treatment. We can visually see from this table that 20C has a higher germination day in both China and the U.S. Thus, we can suspect, that a moderate temperature means a slower germination for the *Ulmus Pumila*. Furthermore, in figure 1 we can see that the boxplots of the two Origins with respect to the Germination Day are identical.

From figure 2 we can see that observations that have been censored only happen at the end of 14 days, thus we can deduce that the censoring variable is fixed and that we've got a type I right censoring dataset. Moreover, we can see that there are some observations that happen on the 14 th day as well, but perhaps the censoring time happened in the afternoon, while these variables were in the morning, thus making this coherent with censoring.

Part B

(a) Visualisation of the Kaplan–Meier estimators

We first visualise the survival distribution using a non-parametric approach.

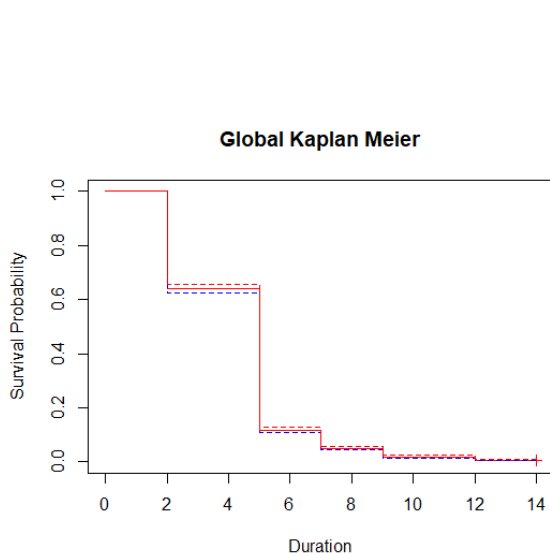


Figure 3: Global Kaplan–Meier estimator

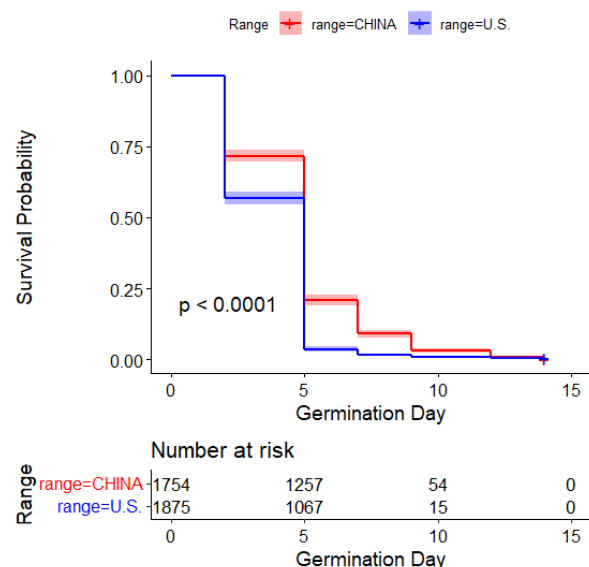


Figure 4: Kaplan–Meier estimators by origin

From Figures 3 and 4, we observe that the survival curves almost reach 0% of survival probability, which is consistent with the low proportion of censored observations in the data.

From Figure 4, the survival function for U.S. seeds is always less than or equal to that of Chinese seeds. This indicates that non-native seeds germinate at least as quickly as native seeds, and in some time intervals, faster.

(b) Quantile estimation and confidence intervals

We now estimate the quantiles of the survival curves together with their confidence intervals. The results are reported in Table 3.

Table 3: Estimates and confidence intervals for survival quantiles

Country	Quantile (%)	Estimate	Lower CI	Upper CI
China	25	2	2	2
China	50	5	5	5
China	75	5	5	5
U.S.	25	2	2	2
U.S.	50	5	5	5
U.S.	75	5	5	5

From Table 3, we observe that the Chinese and U.S. samples yield identical quantile estimates and confidence intervals. Moreover, the confidence intervals collapse to single points and coincide with the point estimates. This is due to the large sample size combined with the discrete nature of the germination times, which take only a small number of possible values, resulting in very tight confidence intervals.

(c) Hypothesis testing

We aim to test whether the survival experiences of the two populations differ. Specifically, we test

$$H_0 : h_{\text{U.S.}}(t) = h_{\text{China}}(t), \forall t \in [0, 14],$$

against

$$H_1 : h_{\text{U.S.}}(t) \neq h_{\text{China}}(t), \text{ for some } t \in [0, 14].$$

Given the non-parametric nature of the problem, we employ weighted log-rank tests. From Figure 4 and the earlier analysis, we observe that most events occur at early times. Therefore, we use the Harrington–Fleming family of tests with $\rho = 0$ (standard log-rank) and $\rho = 1$, placing more emphasis on early differences in survival.

The results of both tests are presented in Table 4.

Table 4: Comparison of log-rank tests with different weighting schemes for germination survival curves

Test (ρ)	Group	N	Observed	Expected	$(O - E)^2/E$	$(O - E)^2/V$
$\rho = 1$	CHINA	1754	1100	1317	35.9	193
	U.S.	1875	1454	1236	38.2	193
	Overall χ^2 (df = 1)					193
	p-value					$< 2 \times 10^{-16}$
$\rho = 0$	CHINA	1754	1747	1999	31.9	200
	U.S.	1875	1864	1612	39.5	200
	Overall χ^2 (df = 1)					200
	p-value					$< 2 \times 10^{-16}$

We observe that both tests lead to the same conclusion: the null hypothesis H_0 is rejected at any conventional significance level. Thus, we conclude that the two survival curves differ significantly. In particular, the estimated hazard for Chinese seeds is higher than that for U.S. seeds, which is consistent with the visual evidence presented in part (a).

(d) Estimation using Cox's model

Remark. In this subsection, we assume that the Cox proportional hazards model is valid. Since the estimated survival curves do not cross, the proportional hazards assumption appears reasonable.

Our objective is to repeat the analysis from part (a) using a semi-parametric Cox proportional hazards model. Specifically, we test (**Hypothesis 1**)

$$H_0 : \beta_{\text{range}} = 0 \quad \text{against} \quad H_1 : \beta_{\text{range}} \neq 0.$$

If $\beta_{\text{range}} = 0$, then the corresponding hazard ratio $\exp(\beta_{\text{range}})$ equals 1, implying that the range has no effect on the hazard of germination and is therefore not statistically significant.

The fitted model is summarized in Table 5.

Table 5: Cox proportional hazards model for germination time

Covariate	Coef.	$\exp(\text{Coef})$	$\text{se}(\text{Coef})$	z	$p\text{-value}$	95% CI (HR)
Range (U.S. vs China)	0.496	1.642	0.034	14.44	$< 2 \times 10^{-16}$	[1.535, 1.756]

Figure 5 displays the survival curves predicted by the Cox model.

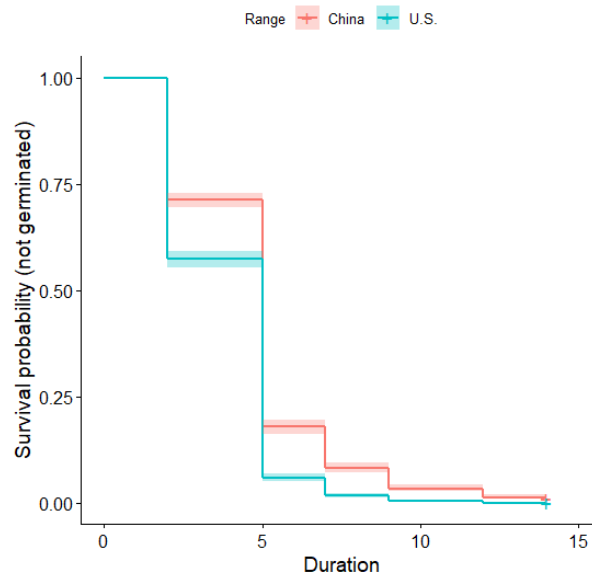


Figure 5: Survival curves predicted by the Cox model

As shown in Figure 5, the fitted survival curves closely resemble the Kaplan–Meier estimates obtained in part (a).

From Table 5, we obtain results consistent with those of part (c). Using the U.S. group as the reference, the estimated hazard ratio is $\exp(\hat{\beta}_{\text{range}}) = 1.642$, indicating that Chinese seeds have a significantly higher hazard of germination than U.S. seeds. The extremely small p -value leads us to reject the null hypothesis $H_0 : \beta_{\text{range}} = 0$, confirming that range has a significant effect on germination time.

Part C

We analyze all relevant variables, excluding `seed` and `population`. The variable `seed` is an identifier with no statistical information, while `population` is nested within `range`, so including both would introduce redundancy and potential multicollinearity. Because our focus is on differences between native and non-native ranges, we retain `range` and exclude `population`.

(a) Relative risk case study

(1) Choosing the model

Based on previously identified relevant variables, we fit a Cox proportional hazards model for germination time:

```
coxfull: Surv(germination_day, status) ~ range + temperature_treatment
          + strata(replicate) + range × temperature_treatment.
```

The variable `temperature_treatment` is included due to its strong influence on germination speed, while `replicate` is used as a stratification factor to control for baseline differences without estimating coefficients. The interaction term allows temperature effects to differ between native and non-native seeds, reflecting potential biological differences in temperature responses.

Model selection is conducted using a Type I (sequential) approach. Given the large sample size, this method is computationally feasible and facilitates comparison of nested models. The candidate models are:

```
m1: Surv(germination_day, status) ~ range + temperature_treatment + range × temperature_treatment,
```

```
m2: Surv(germination_day, status) ~ range + temperature_treatment + strata(replicate).
```

Model comparison based on the Akaike Information Criterion (AIC) is presented in Table 6.

Table 6: Model comparison using Akaike Information Criterion (AIC)

	coxfull	m1	m2
Degrees of freedom	3	3	2
AIC	36242.43	51254.67	36246.50

The full model `coxfull` achieves the lowest AIC, indicating the best trade-off between model fit and complexity among the considered candidates. Consequently, we retain `coxfull` for the remainder of this analysis.

(2) Relative risk estimation and confidence intervals

Having selected the final model, we now estimate the relative risks (hazard ratios) and their associated confidence intervals. The results are reported in Table 7.

Table 7: Estimated hazard ratios from the Cox model

Covariate	Coefficient	Hazard ratio	95% CI	SE	z	p -value
Range (U.S. vs China)	0.169	1.18	(0.85, 1.66)	0.172	**	**
Temperature treatment	0.086	1.09	(1.08, 1.10)	0.0049	17.60	$< 2 \times 10^{-16}$
Range $\times$ Temperature	0.0166	1.02	(1.00, 1.03)	0.0068	2.46	0.0137

All estimated hazard ratios exceed 1, indicating an increased hazard of germination relative to the reference levels, conditional on the other covariates.

In this analysis, inference is based on the parametric coefficient β of the Cox model (see Hypothesis 1). Specifically, the **summary** output of the Cox model reports Wald tests for each coefficient.

The effect of **range** is discussed in greater detail in the following subquestion.

Temperature treatment has a strong and highly significant effect on germination time, with a hazard ratio of 1.09, indicating that higher temperatures substantially accelerate germination. This confirms temperature as the dominant driver of germination dynamics.

The interaction between **range** and **temperature_treatment** is statistically significant. The estimated hazard ratio of 1.02 implies that temperature increases accelerate germination more strongly for non-native seeds than for native seeds. This interaction explains why the marginal effect of **range** becomes non-significant once temperature and its interaction with range are included in the model.

(3) Hypothesis testing on the categorical variable

We now test whether the categorical variable **range** has a statistically significant effect on germination time within the Cox model.

To do so, we perform a likelihood ratio test (LRT), comparing the full Cox model to a reduced model that excludes the **range** main effect. (A Wald test could also be used and would lead to the same conclusion.) The results are reported in Table 8.

Table 8: Likelihood ratio test comparing reduced and full Cox proportional hazards models for time to germination

Model	Log-likelihood	χ^2	df	p -value
Reduced model	-18119	—	—	—
Full model	-18118	0.967	1	0.326

Based on the likelihood ratio test, we fail to reject the null hypothesis. Thus, suggesting that, at the baseline temperature level, germination rates do not differ significantly between U.S. and Chinese seeds. This result contrasts with the findings from Part B, where temperature effects were not explicitly accounted for.

(4) Assessing the proportional hazards assumption

In this subsection, we assess the validity of the proportional hazards (PH) assumption for the covariate **range**. We consider three commonly used approaches: (i) diagnostic plots based on the Nelson–Aalen

estimator, (ii) time-by-covariate interaction diagnostics, and (iii) tests based on Schoenfeld residuals.

We first focus on diagnostic plots derived from the Nelson–Aalen estimator, which are non-parametric and model-independent.

(i) Diagnostic plots

We construct three diagnostic plots:

- Plot 1: $\log H_{\text{China}}(t)$ and $\log H_{\text{U.S.}}(t)$ versus time,
- Plot 2: $\log H_{\text{China}}(t) - \log H_{\text{U.S.}}(t)$ versus time,
- Plot 3: $H_{\text{China}}(t)$ versus $H_{\text{U.S.}}(t)$.

From theoretical results, the PH assumption is supported if: (1) the curves in Plot 1 are approximately parallel, (2) Plot 2 is approximately constant over time, and (3) Plot 3 is approximately linear and passes through the origin.

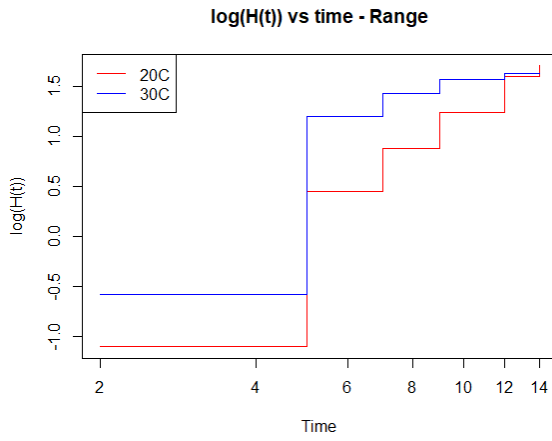


Figure 6: $\log H(t)$ versus time

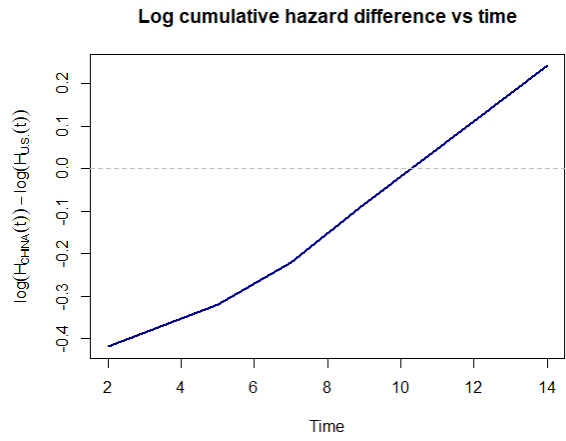


Figure 7: Difference of log cumulative hazards

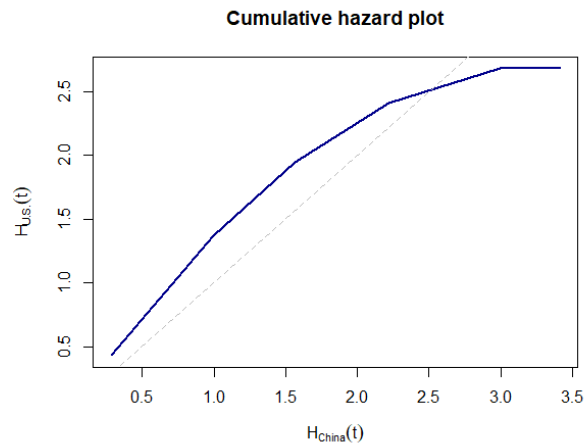


Figure 8: Cumulative hazard for China versus U.S.

None of the expected patterns supporting the PH assumption are clearly observed. In Plot 6, the log

cumulative hazard curves are not parallel, with substantial variation in the distance between the curves over time. In Plot 7, the difference in log cumulative hazards varies markedly at early times before stabilizing, indicating time-dependent effects. In Plot 8, the relationship between cumulative hazards is clearly nonlinear, especially at early times.

Overall, these diagnostics suggest a violation of the proportional hazards assumption, particularly during the early phase of the germination process. However, it is important to note that these diagnostics are visual and non model-based. Consequently, while they raise concerns about the PH assumption, they should be complemented with formal, model-based tests before drawing a definitive conclusion.

Time-by-covariate interaction diagnostic

This approach allows us to assess the proportional hazards (PH) assumption by explicitly modeling time-dependent effects. In addition, it provides insight into how the hazard ratio evolves over time.

Since the covariate **range** is binary, we define a time-dependent function

$$g(t) = \mathbb{I}(t \leq 5),$$

which allows the effect of **range** to differ between early and late germination times. The resulting Cox model includes a time-by-covariate interaction term. The estimated coefficients are reported in Table 9.

Table 9: Cox proportional hazards model with time-dependent effects

Covariate	Coefficient	Hazard ratio	SE	z	p -value
Range (U.S.)	-0.528	0.590	0.219	-2.41	0.016
Temperature treatment	0.089	1.093	0.0049	17.95	$< 2 \times 10^{-16}$
tt(range)	0.866	2.378	0.158	5.47	4.42×10^{-8}
Range (U.S.) $\times$ Temperature	0.012	1.012	0.0068	1.79	0.074

The coefficient associated with **tt(range)** is highly significant, providing strong evidence of a time-dependent effect of **range**. This indicates that the hazard ratio between U.S. and Chinese seeds is not constant over time, thereby violating the proportional hazards assumption. Consequently, this diagnostic confirms the conclusions drawn from the Nelson–Aalen diagnostic plots.

Schoenfeld residuals

We now formally test the proportional hazards assumption using Schoenfeld residuals. The null hypothesis for each covariate is that its corresponding Schoenfeld residuals are independent of time, which implies proportional hazards. The alternative hypothesis is that the residuals depend on time, indicating a violation of the PH assumption.

The test results are presented in Table 10, and the residual plots are shown in Figure 9.

Table 10: Tests of proportional hazards using Schoenfeld residuals

Effect	χ^2	df	p -value
Range	7.55	1	0.006
Temperature treatment	428.62	1	$< 2 \times 10^{-16}$
Range $\times$ Temperature	51.41	1	7.5×10^{-13}
Global	473.63	3	$< 2 \times 10^{-16}$

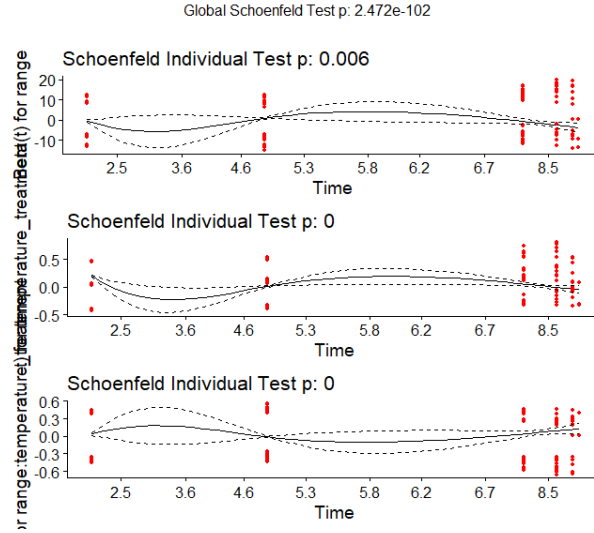


Figure 9: Schoenfeld residuals versus time

The Schoenfeld residual tests indicate significant time dependence for all covariates, including **range**. In particular, the global test strongly rejects the null hypothesis of proportional hazards. These results provide formal statistical evidence that the PH assumption is violated for the fitted Cox model, which is consistent with the conclusions obtained from both the Nelson–Aalen diagnostics and the time-by-covariate interaction analysis.

Conclusion - (4)

Taken together, these diagnostic procedures provide strong evidence against the proportional hazards (PH) assumption for the covariate **range**. Both graphical diagnostics based on the Nelson–Aalen estimator and the time-by-covariate interaction approach clearly indicate a violation of the PH assumption. Although the test based on Schoenfeld residuals yields a borderline result, it does not provide sufficient support for the validity of the assumption. Consequently, we conclude that the proportional hazards assumption is not satisfied for **range**. As a result, the Cox proportional hazards model is not appropriate in this setting, and any previously derived conclusions based on this model should be interpreted with caution and should not be relied upon for substantive data analysis.

(b) – Parametric regression models

Since the proportional hazards (PH) assumption is violated, we now consider modeling the germination time using a parametric survival model.

(1) – Construction of a parametric model

For consistency, we retain the same set of covariates as in the previous model. The main remaining choice concerns the distribution of survival times. From the earlier analysis, we observed that the PH assumption is not satisfied and that a large proportion of events occur early in time. Consequently, exponential and Weibull distributions—which impose strong assumptions on the hazard function—are not well suited to the data.

It is therefore preferable to consider parametric distributions that allow for non-monotone or time-varying hazards and do not rely on the PH assumption. Two common alternatives are the log-normal and log-

logistic distributions. Model comparison based on the Akaike Information Criterion (AIC) yields

$$AIC_{\text{log-logistic}} = 13157.04 \quad \text{and} \quad AIC_{\text{log-normal}} = 13298.79,$$

with both models having six degrees of freedom. Based on these results, we select the log-logistic model.

(2) – Point and interval estimation of the model coefficients

We recall that the coefficient estimates reported by `survreg` in R correspond to the log-linear representation of the model rather than the accelerated failure time (AFT) parameterization. To interpret the effects in terms of the AFT model, the estimated coefficients must therefore be multiplied by -1 .

Table 11: Log-logistic accelerated failure time (AFT) model for time to germination.

Covariate	Estimate	95% CI	Std. Error	z	p -value
Intercept	2.980	[2.882, 3.079]	0.050	59.44	$< 2 \times 10^{-16}$
Range (U.S.)	0.123	[0.000, 0.245]	0.063	1.97	0.049
Temperature treatment	-0.064	[-0.068, -0.060]	0.002	-30.89	$< 2 \times 10^{-16}$
Replicate	0.004	[-0.001, 0.009]	0.003	1.41	0.160
Range (U.S.) $\times$ Temperature	-0.013	[-0.018, -0.008]	0.003	-5.03	5×10^{-7}

Scale parameter = 0.215

The log-logistic accelerated failure time model indicates strong effects of temperature and geographic range on time to germination (Table 11). Higher incubation temperature significantly accelerated germination, with seeds at higher temperatures germinating substantially earlier than those at lower temperatures (estimate = 0.064, $p < 0.001$). Seeds originating from the U.S. tended to germinate earlier than those from the native range, although this effect was weak and only marginally significant (estimate = 0.123, $p = 0.049$).

A significant interaction between range and temperature indicates that the accelerating effect of temperature on germination was reduced for U.S. populations relative to native populations (estimate = 0.013, $p < 0.001$), suggesting differential thermal responses between ranges.

Overall, the results show that temperature is the primary driver of germination timing, while geographic origin modulates the magnitude of this response.

(3)- Comparison between the parametric model and cox model result

Results from the Cox proportional hazards and log-logistic accelerated failure time models were qualitatively consistent for temperature, with higher temperatures strongly accelerating germination in both approaches. However, effect sizes and statistical significance differed for geographic range and its interaction with temperature. The Cox model did not detect a significant main effect of range, whereas the log-logistic model indicated slightly earlier germination for U.S. populations. Overall, the parametric model provides a more appropriate description of germination timing while supporting the same biological conclusion regarding the dominant role of temperature.

Conclusion

This project examined whether U.S. (non-native) populations germinate faster than Chinese populations using survival analysis. Kaplan–Meier curves and the Harrington–Fleming test indicated clear differences in germination dynamics, with Chinese populations showing higher survival probabilities over time, consistent with slower germination compared to U.S. populations.

Cox model analyses suggested differences in hazards between ranges and highlighted temperature as a highly significant driver of germination. Although range alone was not consistently significant, the interaction between range and temperature was, indicating that population origin modifies the effect of temperature on germination. However, violations of the proportional hazards assumption limited the reliability of Cox-based inference.

A parametric log-logistic model provided a better fit and confirmed that temperature and the range–temperature interaction are the most important predictors, with range showing borderline significance. Overall, the results suggest that U.S. populations tend to germinate faster than Chinese populations, but this advantage depends strongly on temperature conditions.